

John Marsden

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[LinkedIn](#) | [GitHub \(Professional\)](#) | [GitHub \(Personal\)](#)

EDUCATION

North Carolina State University

M.S. – Computer Science

Graduation Date: 04/2025

Abilene Christian University

B.S. – Math and Computer Science

Graduation Date: 04/2020

SKILLS

Databases: MySQL, Oracle Database, Postgres

Dev Ops: Bash, Caddy, DigitalOcean, Docker, Github Actions, Heroku, Linux, Nginx, Redis

Frameworks: ASP.NET, Astro, AsyncAPI, Django, Flask, GraphQL, jQuery, Node.js, Puppeteer, Rails, Ruby on Rails, Selenium, Tornado, Vue

Machine Learning: Attention Networks, Graph Neural Networks, Keras, Matplotlib, Natural Language Processing, Neural Networks, Numpy, Pandas, Pytorch, R, Scikit-learn, Spacy, Tensorflow

Mobile Development: Android, Xamarin

Programming Languages: C, C++, C#, CSS, F#, Haskell, HTML, Java, JavaScript, Kotlin, Lua, PHP, Prolog, Python, Ruby, Rust, SQL, Typescript

Teaching: Curriculum Development, Lesson Planning, Student Assessment, Flipped Classroom, Universal Design for Learning

Instructional Technology: Learning Management Systems, Canvas, Moodle

Testing & Debugging: Cucumber, Fuzzing, GDB, Gherkin, JUnit, Maven, Unit & Integration Testing

Theory: Cryptography, Data Structures & Algorithms, Discrete Mathematics

EXPERIENCE

PhD Student | [AIAL](#)

05/2021 – 11/2024

- Implemented a novel intermediate language for representing student code for use by **graph-based neural networks**
- Performed a comprehensive systematic literature review of 500+ papers from the **debugging education** literature
- Designed 3 studies for use by our research lab to investigate a variety of methods for better teaching and learning debugging

Teaching Assistant | [NCSU](#)

07/2020 – 11/2024

- Helped 300+ students navigate the complexities of multi-project **Java** codebases
- Graded 400+ student algorithms using common data structures, including stacks, graphs, etc.

Instructor - Software Development Fundamentals | [NCSU](#)

05/2022 – 07/2022

- Taught the software development process and **test-driven development** to 20 students
- Demonstrated idiomatic generic object-oriented **Java** code to students during 5+ live coding sessions

Instructor - Introduction to Computing | [NCSU](#)

12/2021 – 04/2022

- Taught the fundamentals of **Java** to 40 students
- Built two autograders using **JUnit 5** to autograde student code submissions

Software Development Intern | [Secmation](#)

04/2021 – 07/2021

- Planned and implemented 20% of a backend for a programming environment with **Python**
- Outlined an API for the programming environment backend using **AsyncAPI**
- Containerized the programming environment backend using **Docker** for scalability and ease of use

Teaching Assistant | [Abilene Christian University](#)

07/2019 – 04/2020

- Designed and implemented 2 interesting and relevant auto-graded programming problems for the object-oriented programming class
- Updated current programming problems to reflect new standards for the automated assessment tool
- Designed and implemented proof-of-concept C# auto-grader for Athene

Report Specialist | [ACU](#)

09/2017 – 04/2020

- Crafted SQL queries to aggregate raw data about alumni interaction into usable information for reporting
- Built Cognos reports to present data in a user-friendly format for relevant stakeholders

Software Development Intern | [Faithlife](#)

04/2019 – 07/2019

- Converted the **OAuth** application management portal from being user specific to a group-based system
- Integrated a **HaveIBeenPwned** check using the **k-anonymity** API

Tutor | [ACU](#)

07/2017 – 04/2019

- Tutored 70+ students on **C++** programming concepts, including the basics, pointers, memory management, and object-oriented programming
- Graded 200+ student **C++** algorithm implementations using common data structures, including queues, graphs, trees, etc.

PROJECTS